Beginner's Guide to Pen Plotter - Generating G Code with inkscape

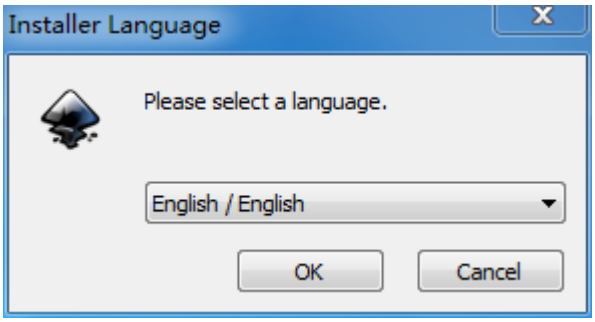
1. Prepare the software

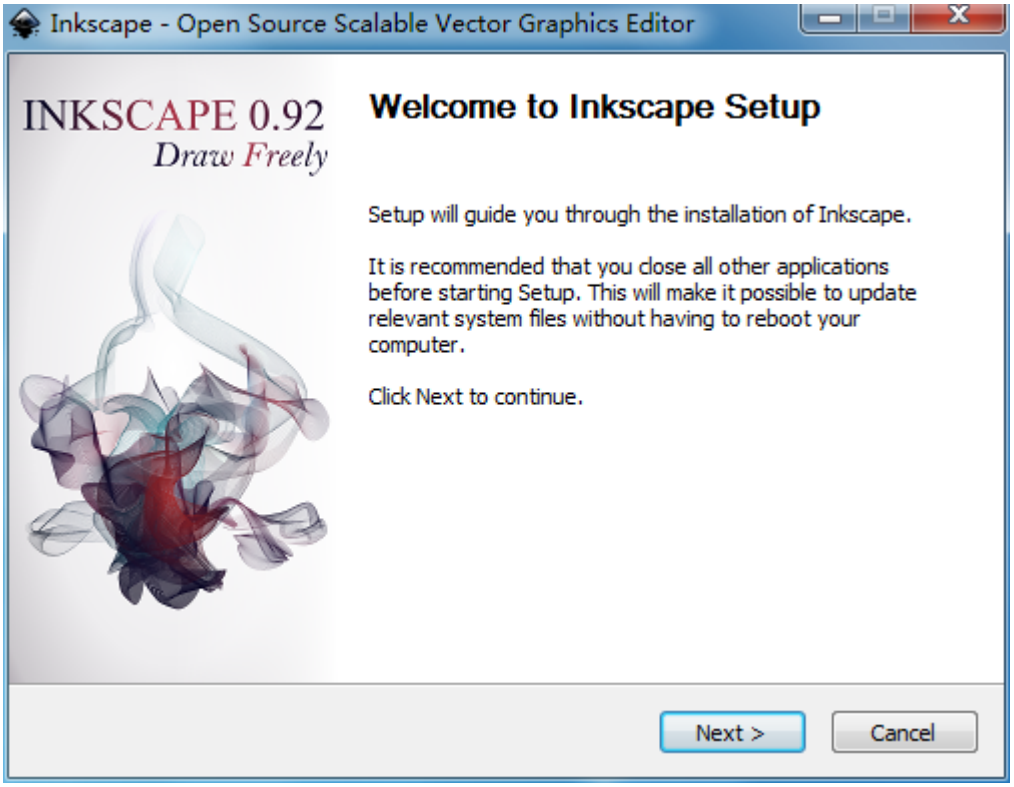
https://universalgcodesender.com/download

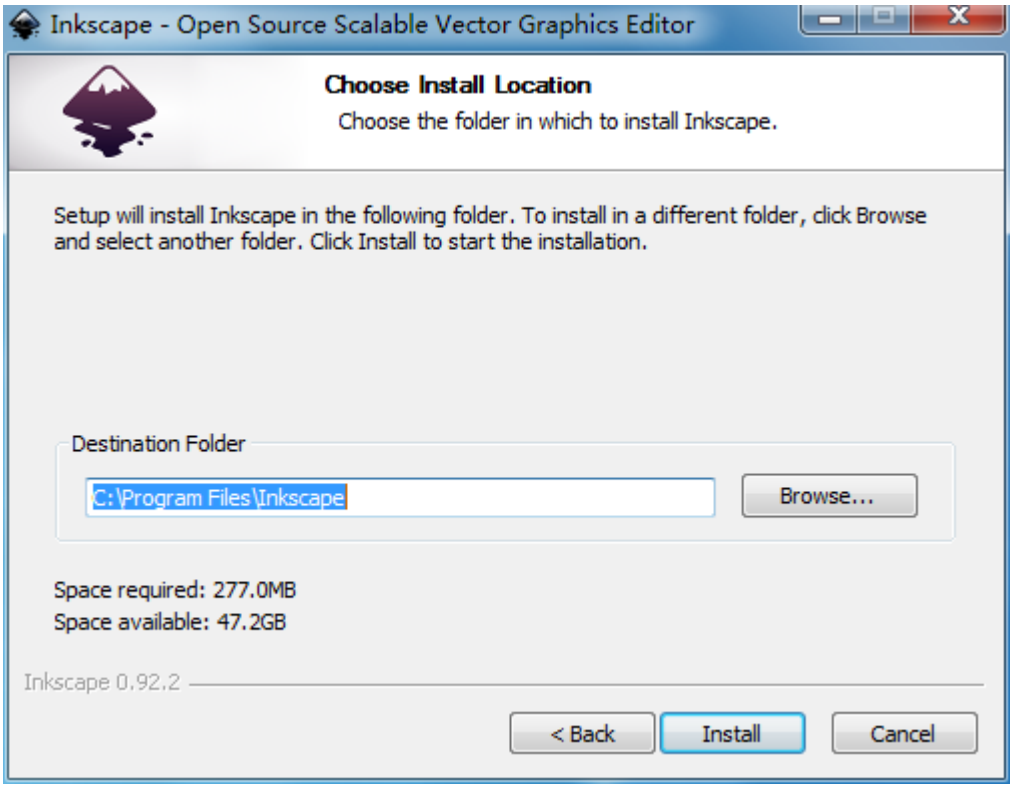
https://github.com/Denvi/Candle/releases/tag/v0.9

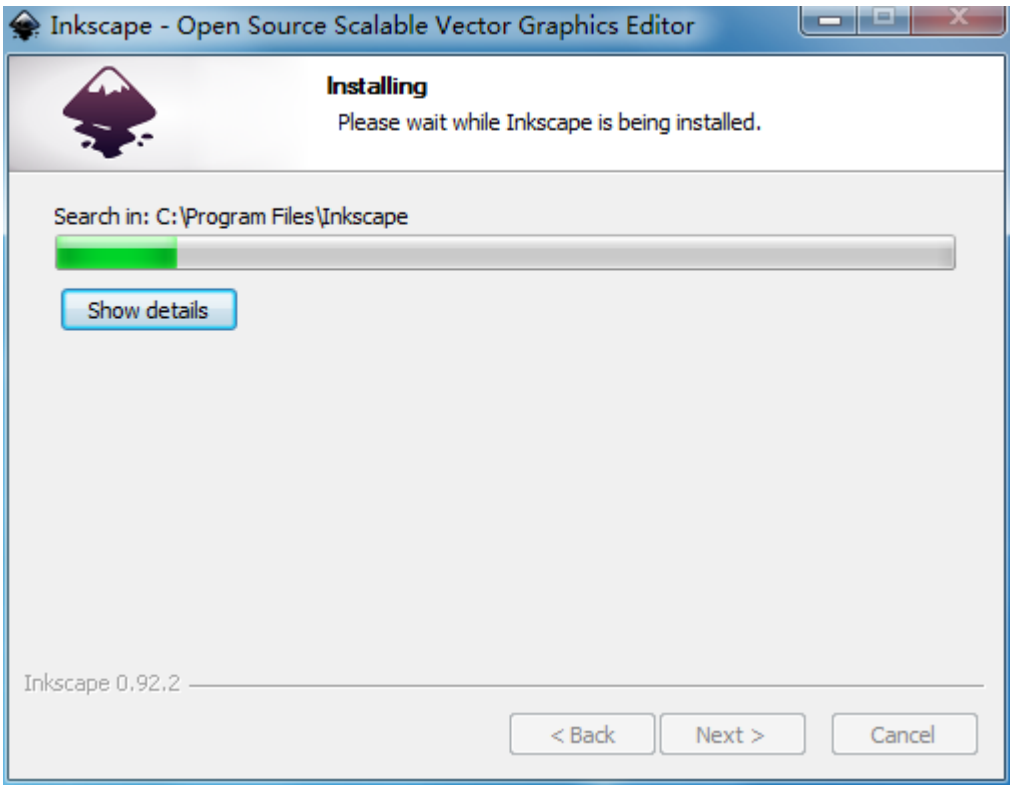
https://inkscape.org/release/inkscape-0.92.2/

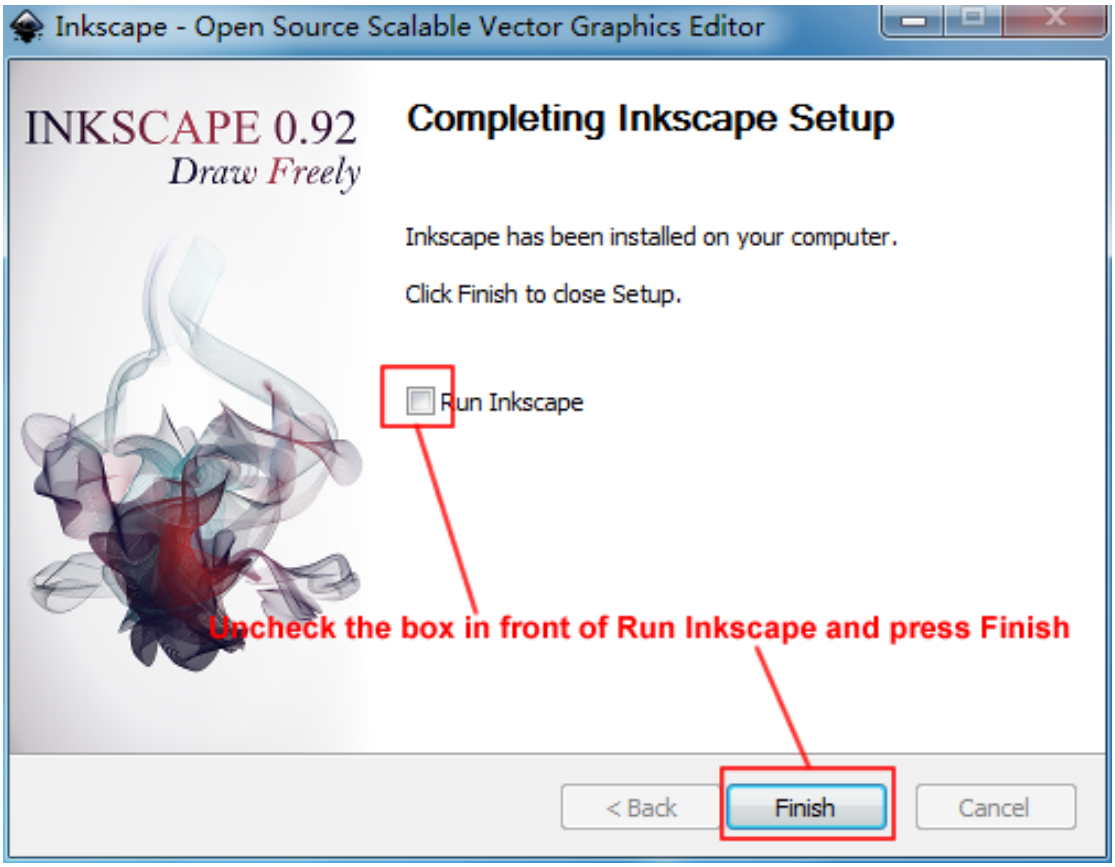
2. Install inkscape software and plugins



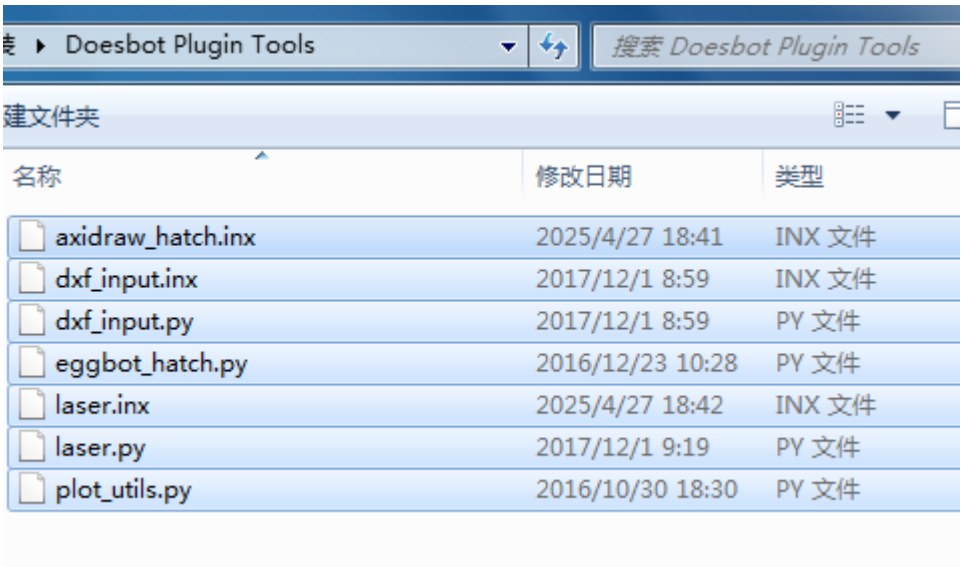


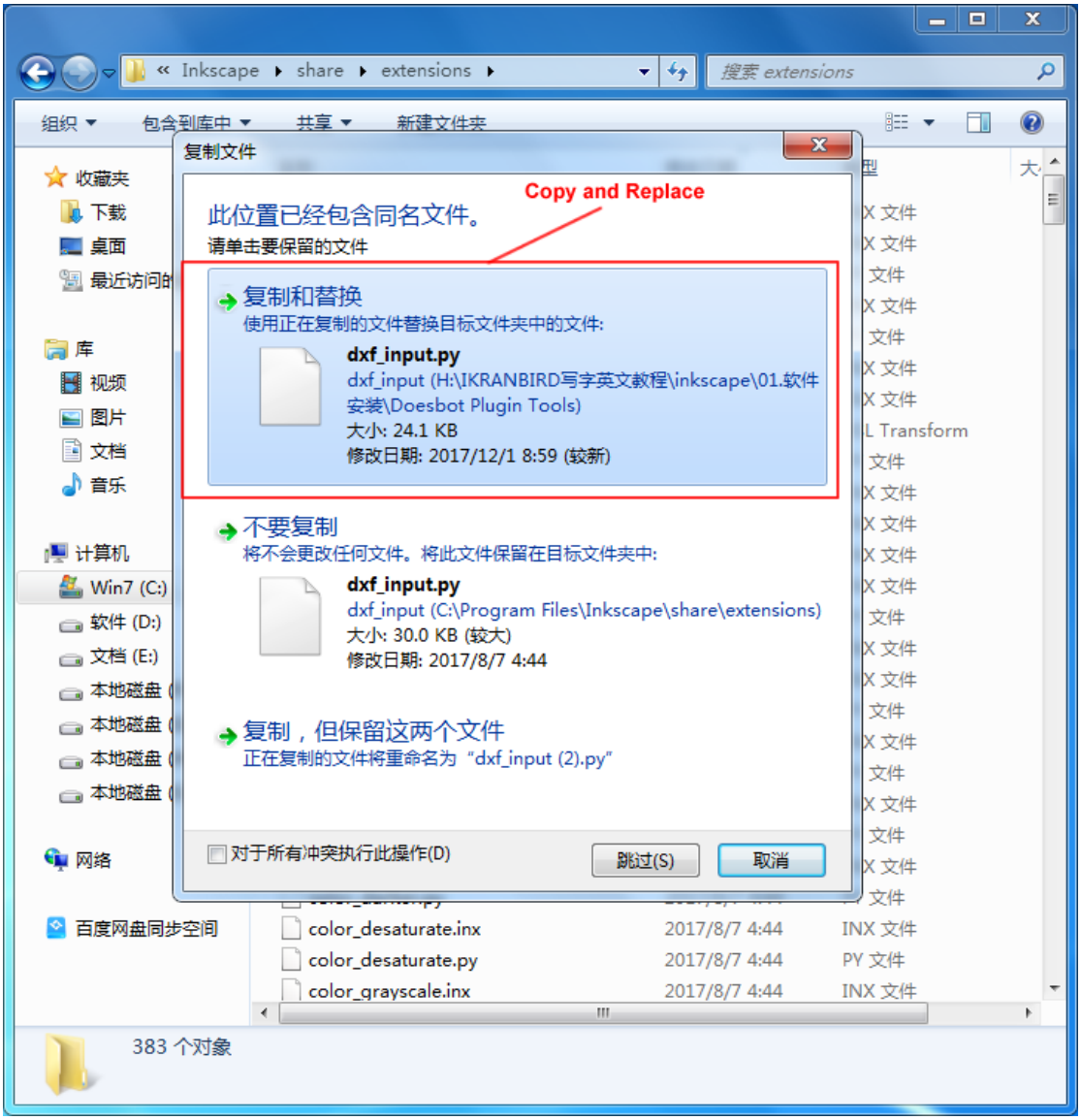

Inkscape - Open Source Scalable Vector Graphics Editor
Choose Install Location
Choose the folder in which to install Inkscape.
Setup will install Inkscape in the following folder. To install in a different folder, click Browse and select another folder. Click Install to start the installation.
Destination Folder
C:\Program Files\Inkscape
Browse...
Space required: 277.0MB
Space available: 47.2GB
Inkscape 0.92.2
< Back
Install
Cancel


Inkscape - Open Source Scalable Vector Graphics Editor
Installing
Please wait while Inkscape is being installed.
Search in: C:\Program Files\Inkscape
Show details
Inkscape 0.92.2
< Back
Next >
Cancel

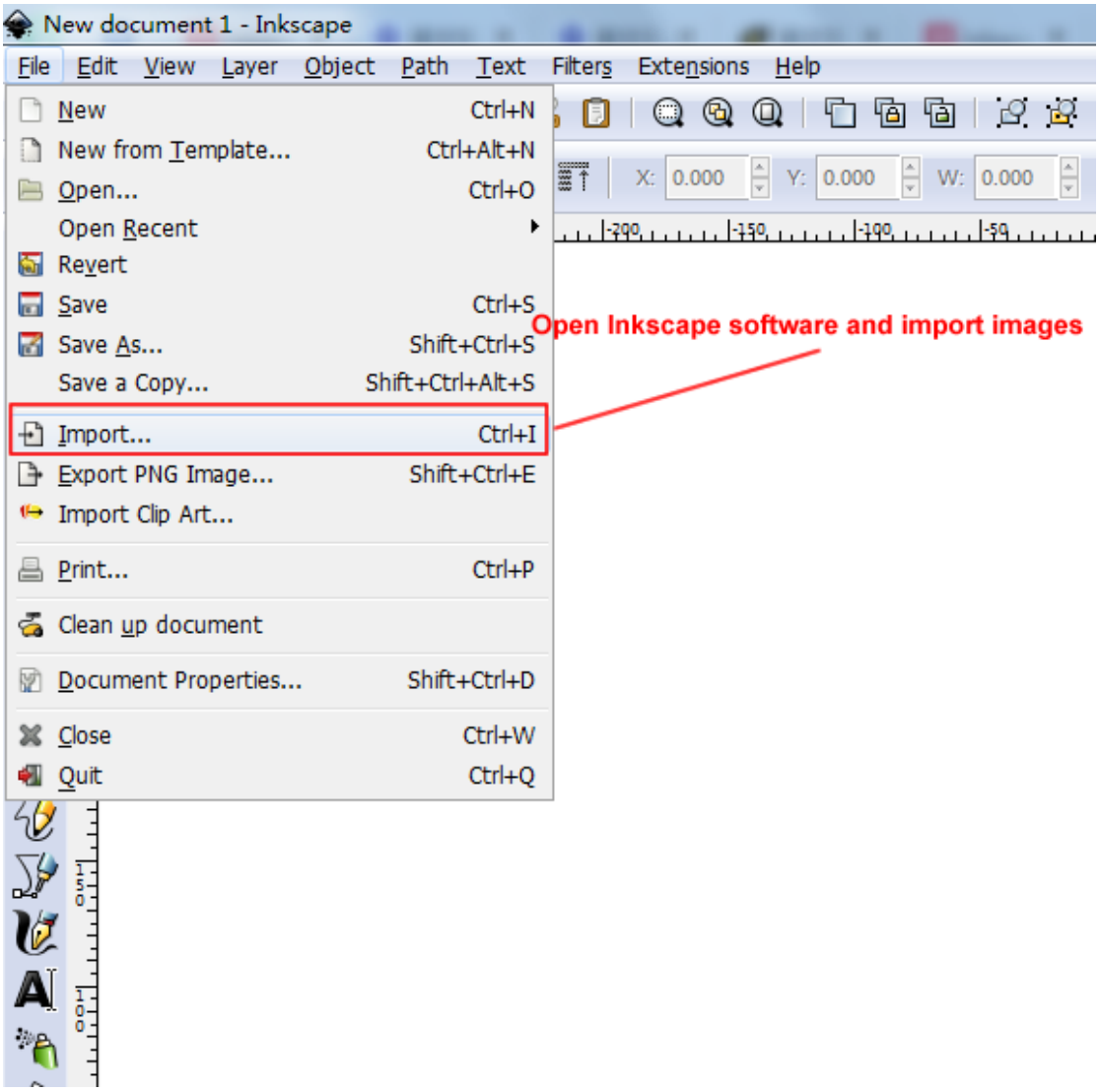


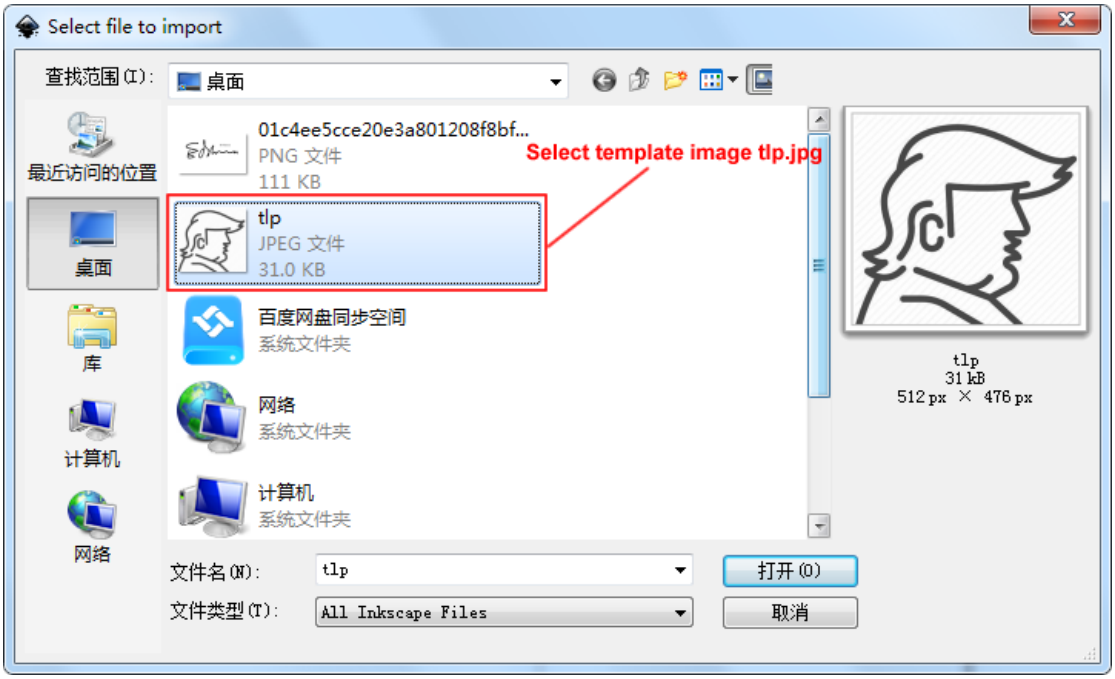
Install the plugin and copy 7 files from the Doesbot Plugin Tools folder to the installation directory/inkscape/share/extensions/

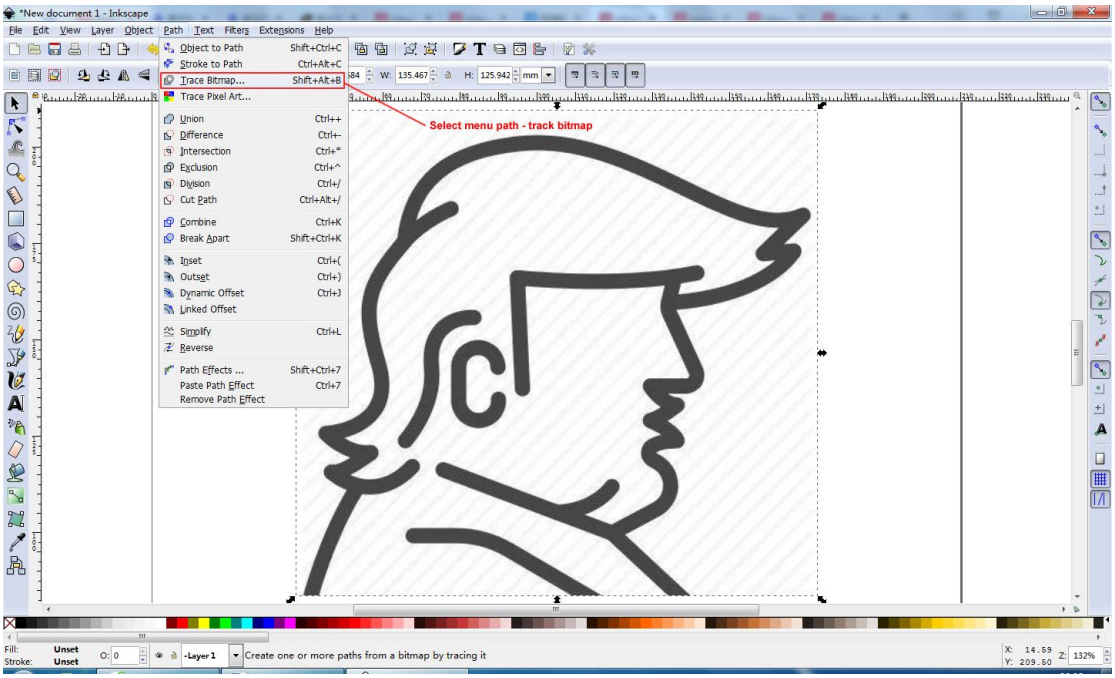


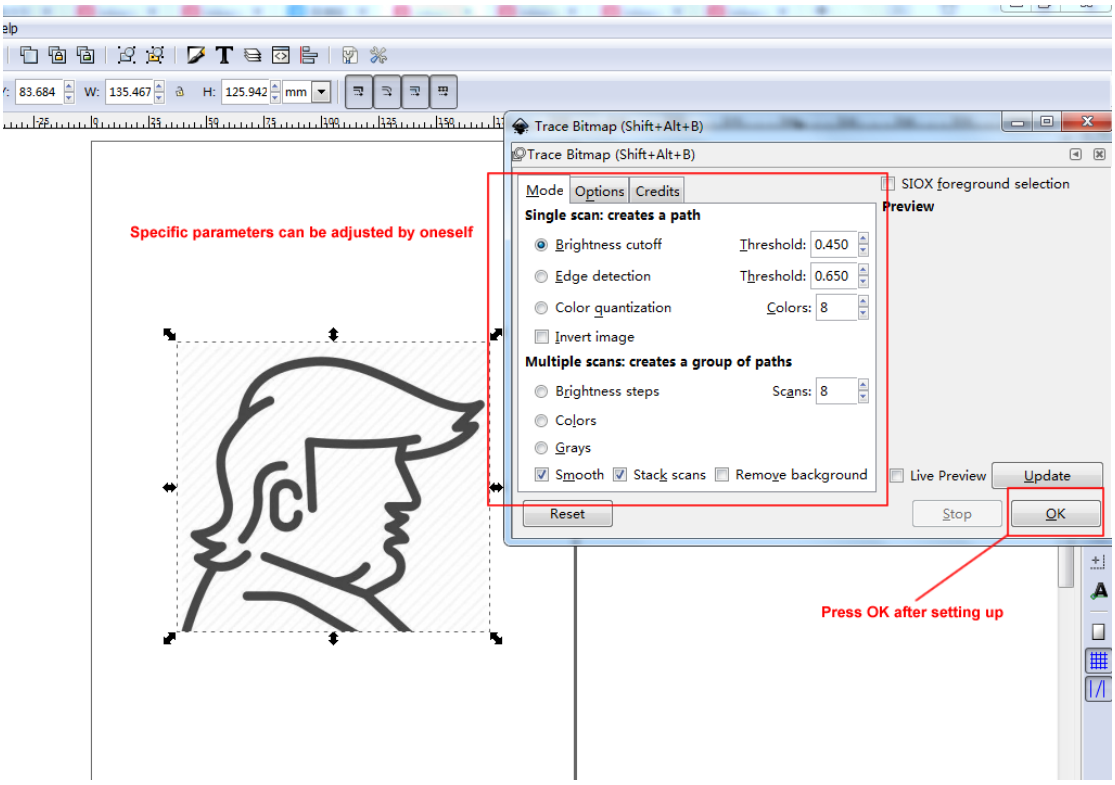

« Inkscape ▸ share ▸ extensions ▸
搜索 extensions
组织
包含到库中
共享
新建文件夹
复制文件
Copy and Replace
此位置已经包含同名文件。
请单击要保留的文件
复制和替换
使用正在复制的文件替换目标文件夹中的文件:
dxf_input.py
dxf_input (H:\IKRANBIRD写字英文教程\inkscape\01.软件安装\Doesbot Plugin Tools)
大小: 24.1 KB
修改日期: 2017/12/1 8:59 (较新)
不要复制
将不会更改任何文件。将此文件保留在目标文件夹中:
dxf_input.py
dxf_input (C:\Program Files\Inkscape\share\extensions)
大小: 30.0 KB (较大)
修改日期: 2017/8/7 4:44
复制，但保留这两个文件
正在复制的文件将重命名为 "dxf_input (2).py"
对于所有冲突执行此操作(D)
跳过(S)
取消
收藏夹
下载
桌面
最近访问的
库
视频
图片
文档
音乐
计算机
Win7 (C:)
软件 (D:)
文档 (E:)
网络
百度网盘同步空间
color_desaturate.inx 2017/8/7 4:44 INX 文件
color_desaturate.py 2017/8/7 4:44 PY 文件
color_grayscale.inx 2017/8/7 4:44 INX 文件
383 个对象

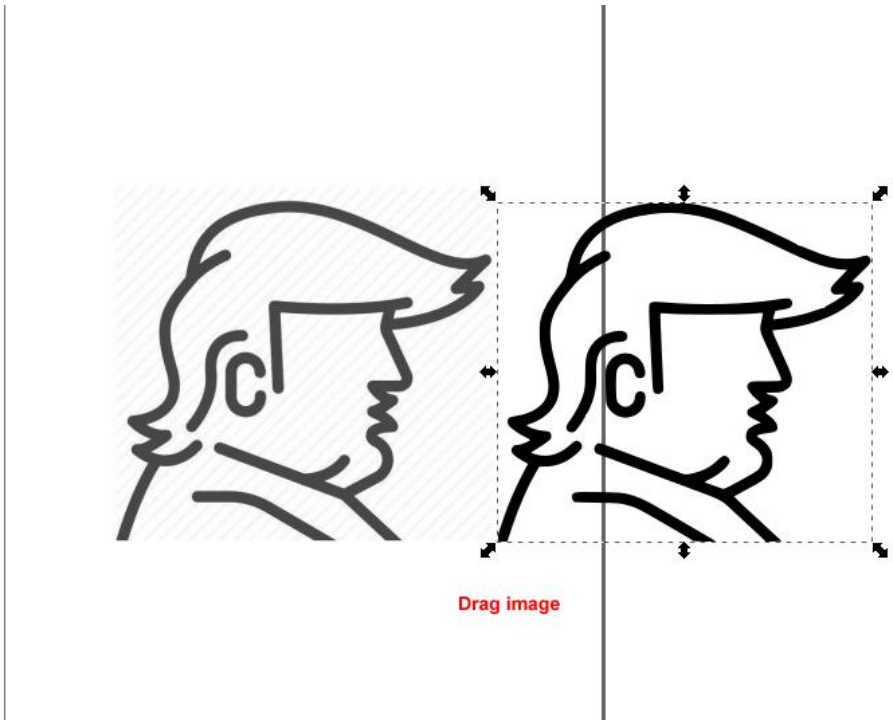
3. Image G code generation

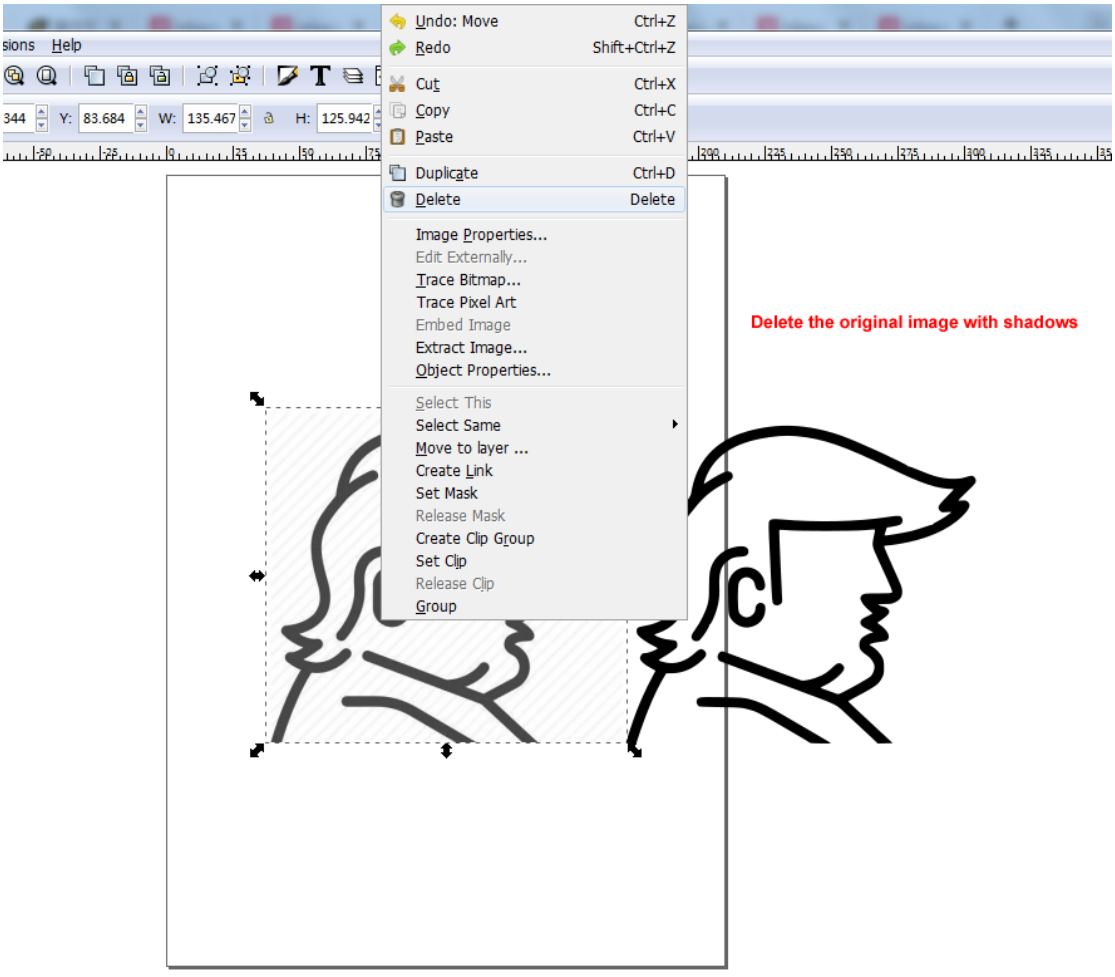


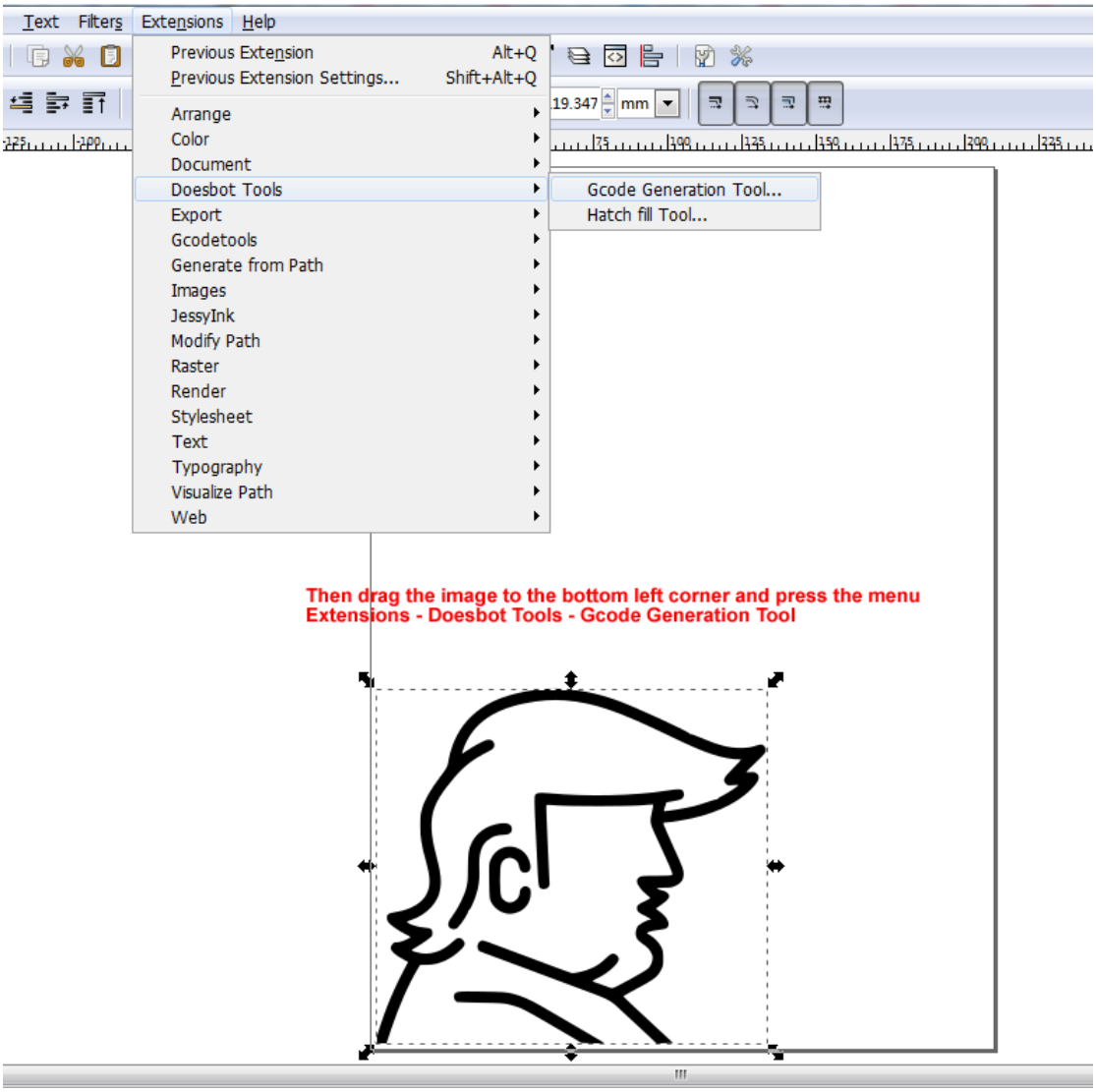


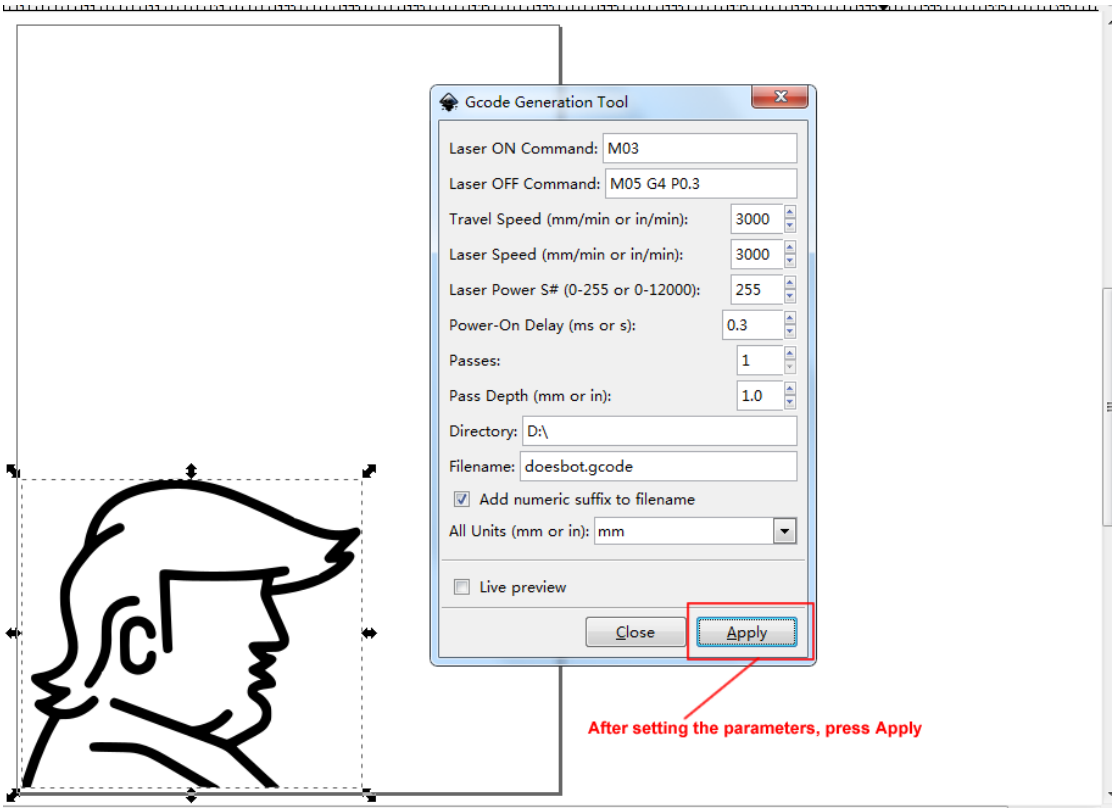

*New document 1 - Inkscape
File Edit View Layer Object Path Text Filters Extensions Help
Object to Path Shift+Ctrl+C
Stroke to Path Ctrl+Alt+C
Trace Bitmap... Shift+Alt+B
Trace Pixel Art...
Union Ctrl++
Difference Ctrl+-
Intersection Ctrl+*
Exclusion Ctrl+^
Division Ctrl+/
Cut Path Ctrl+Alt+/
Combine Ctrl+K
Break Apart Shift+Ctrl+K
Inset Ctrl+(
Outset Ctrl+)
Dynamic Offset Ctrl+J
Linked Offset
Simplify Ctrl+L
Reverse
Path Effects... Shift+Ctrl+7
Paste Path Effect Ctrl+7
Remove Path Effect
Select menu path - track bitmap
Create one or more paths from a bitmap by tracing it

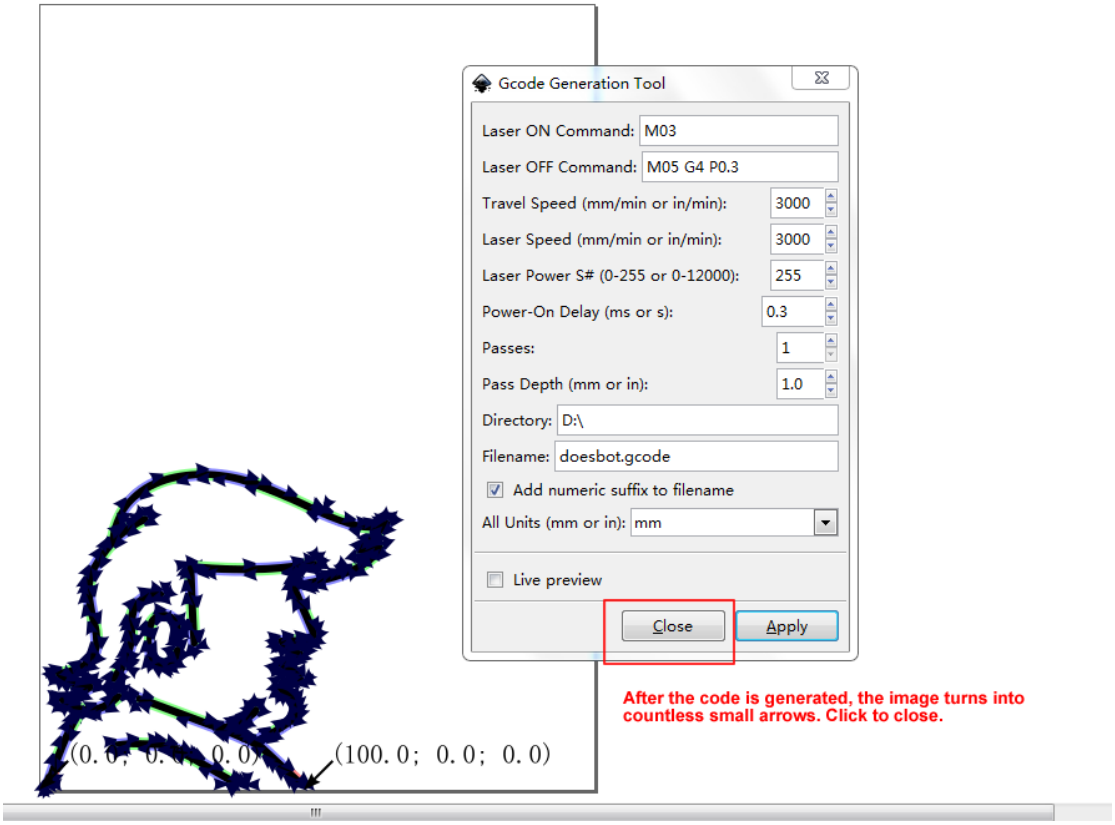

Trace Bitmap (Shift+Alt+B)
Mode Options Credits
SIOX foreground selection
Preview
Single scan: creates a path
Brightness cutoff Threshold: 0.450
Edge detection Threshold: 0.650
Color quantization Colors: 8
Invert image
Multiple scans: creates a group of paths
Brightness steps Scans: 8
Colors
Grays
Smooth Stack scans Remove background
Live Preview Update
Reset Stop OK
Specific parameters can be adjusted by oneself
Press OK after setting up

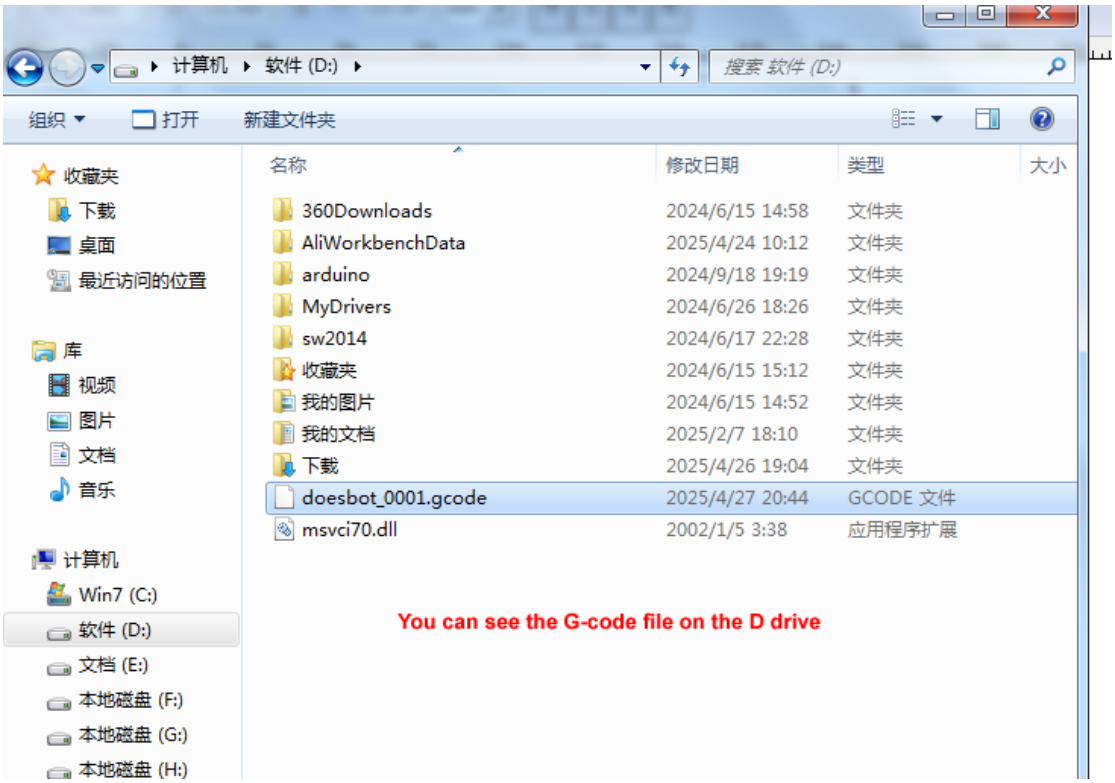

Drag image

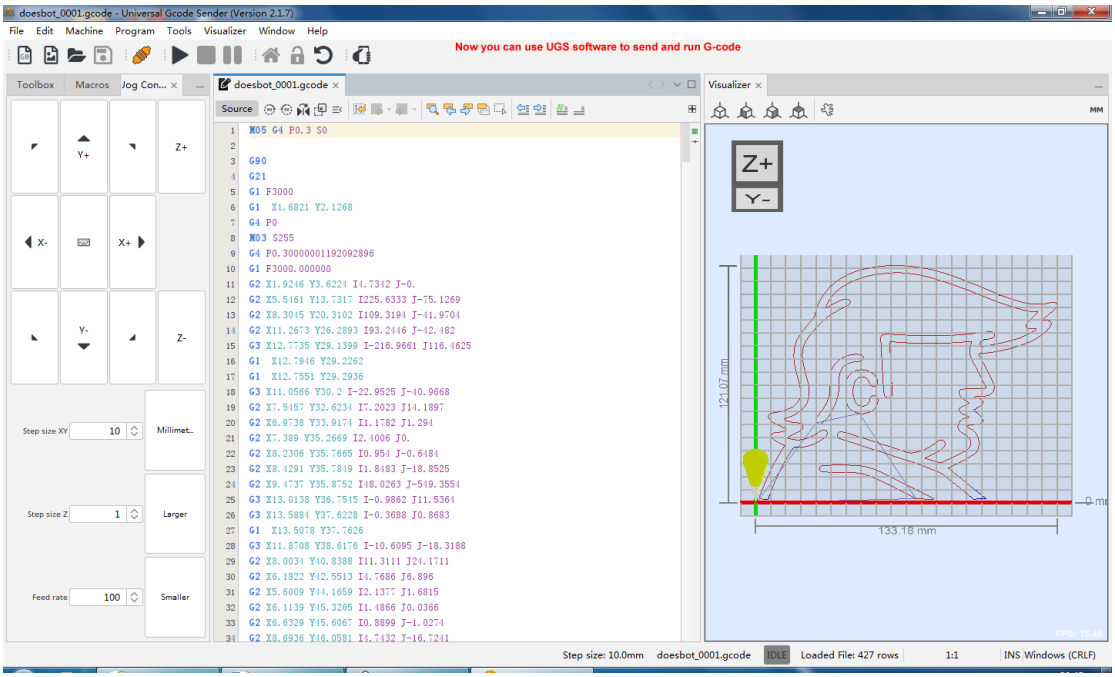

sions Help
344 Y: 83.684 W: 135.467 H: 125.942
Undo: Move Ctrl+Z
Redo Shift+Ctrl+Z
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Duplicate Ctrl+D
Delete Delete
Image Properties...
Edit Externally...
Trace Bitmap...
Trace Pixel Art
Embed Image
Extract Image...
Object Properties...
Select This
Select Same
Move to layer ...
Create Link
Set Mask
Release Mask
Create Clip Group
Set Clip
Release Clip
Group
Delete the original image with shadows


Text
Filters
Extensions
Help
Previous Extension Alt+Q
Previous Extension Settings... Shift+Alt+Q
Arrange
Color
Document
Doesbot Tools
Export
Gcodetools
Generate from Path
Images
JessyInk
Modify Path
Raster
Render
Stylesheet
Text
Typography
Visualize Path
Web
Gcode Generation Tool...
Hatch fill Tool...
19.347
mm
Then drag the image to the bottom left corner and press the menu
Extensions - Doesbot Tools - Gcode Generation Tool

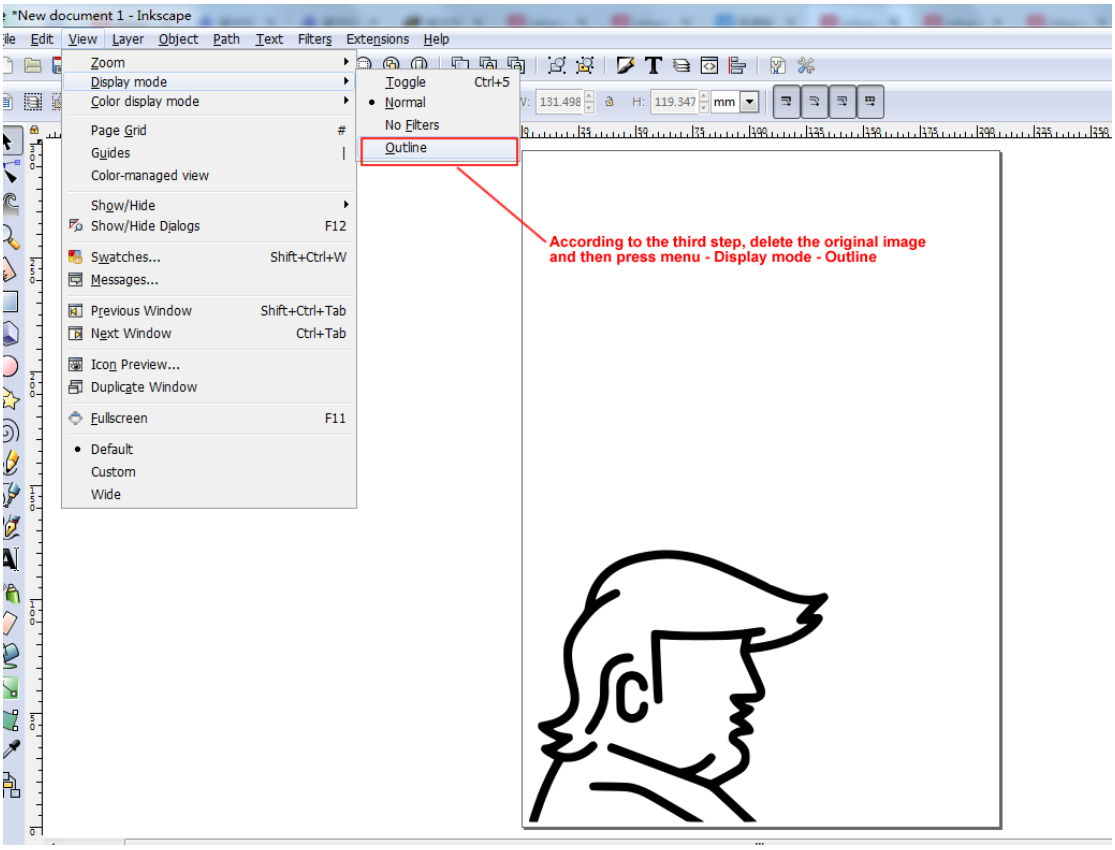

Gcode Generation Tool
Laser ON Command: M03
Laser OFF Command: M05 G4 P0.3
Travel Speed (mm/min or in/min): 3000
Laser Speed (mm/min or in/min): 3000
Laser Power S# (0-255 or 0-12000): 255
Power-On Delay (ms or s): 0.3
Passes: 1
Pass Depth (mm or in): 1.0
Directory: D:\
Filename: doesbot.gcode
Add numeric suffix to filename
All Units (mm or in): mm
Live preview
Close
Apply
After setting the parameters, press Apply

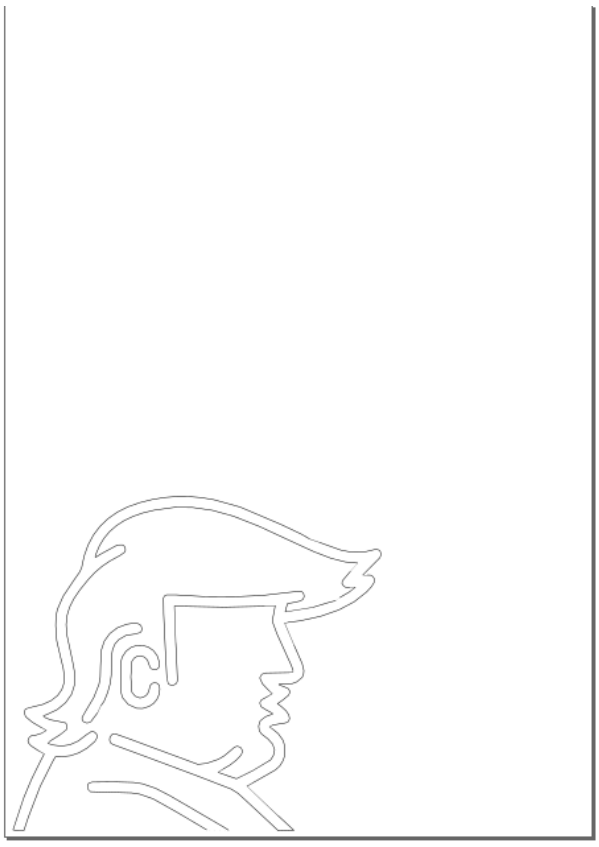

Gcode Generation Tool
Laser ON Command: M03
Laser OFF Command: M05 G4 P0.3
Travel Speed (mm/min or in/min): 3000
Laser Speed (mm/min or in/min): 3000
Laser Power S# (0-255 or 0-12000): 255
Power-On Delay (ms or s): 0.3
Passes: 1
Pass Depth (mm or in): 1.0
Directory: D:\
Filename: doesbot.gcode
Add numeric suffix to filename
All Units (mm or in): mm
Live preview
Close
Apply
After the code is generated, the image turns into countless small arrows. Click to close.
(100.0; 0.0; 0.0)

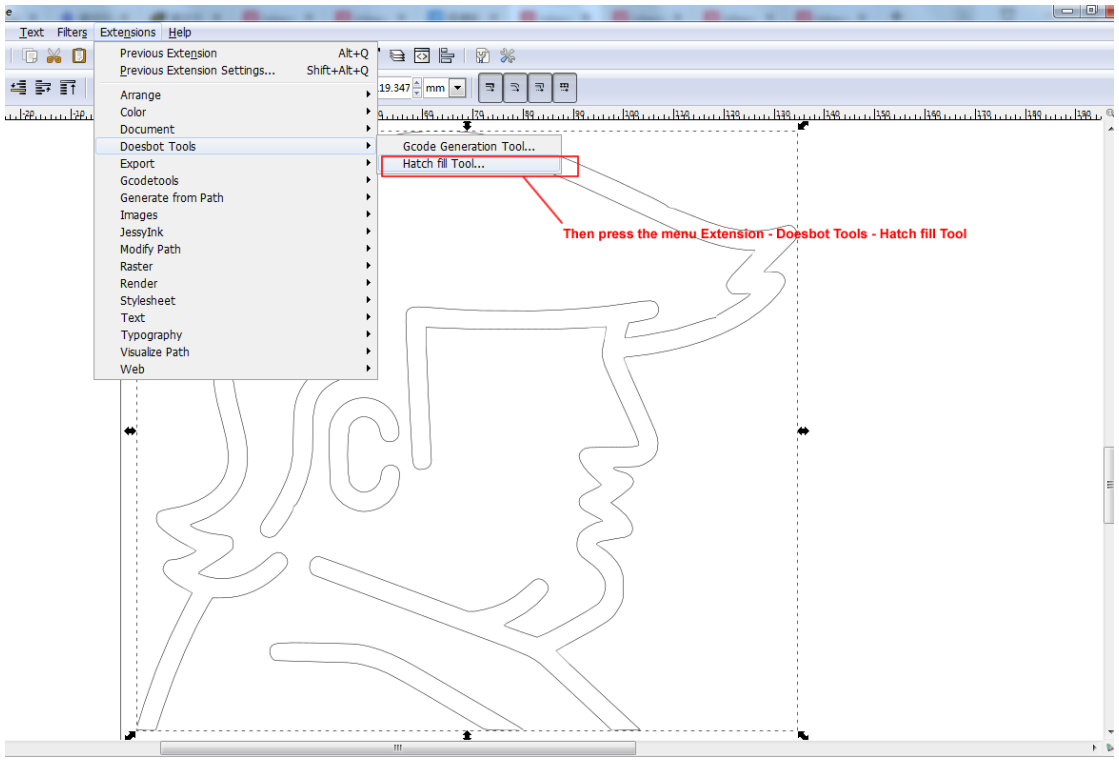

计算机 ▸ 软件 (D:) ▸
搜索 软件 (D:)
组织
打开
新建文件夹
收藏夹
下载
桌面
最近访问的位置
库
视频
图片
文档
音乐
计算机
Win7 (C:)
软件 (D:)
文档 (E:)
本地磁盘 (F:)
本地磁盘 (G:)
本地磁盘 (H:)
名称
修改日期
类型
大小
360Downloads 2024/6/15 14:58 文件夹
AliWorkbenchData 2025/4/24 10:12 文件夹
arduino 2024/9/18 19:19 文件夹
MyDrivers 2024/6/26 18:26 文件夹
sw2014 2024/6/17 22:28 文件夹
收藏夹 2024/6/15 15:12 文件夹
我的图片 2024/6/15 14:52 文件夹
我的文档 2025/2/7 18:10 文件夹
下载 2025/4/26 19:04 文件夹
doesbot_0001.gcode 2025/4/27 20:44 GCODE 文件
msvci70.dll 2002/1/5 3:38 应用程序扩展
You can see the G-code file on the D drive

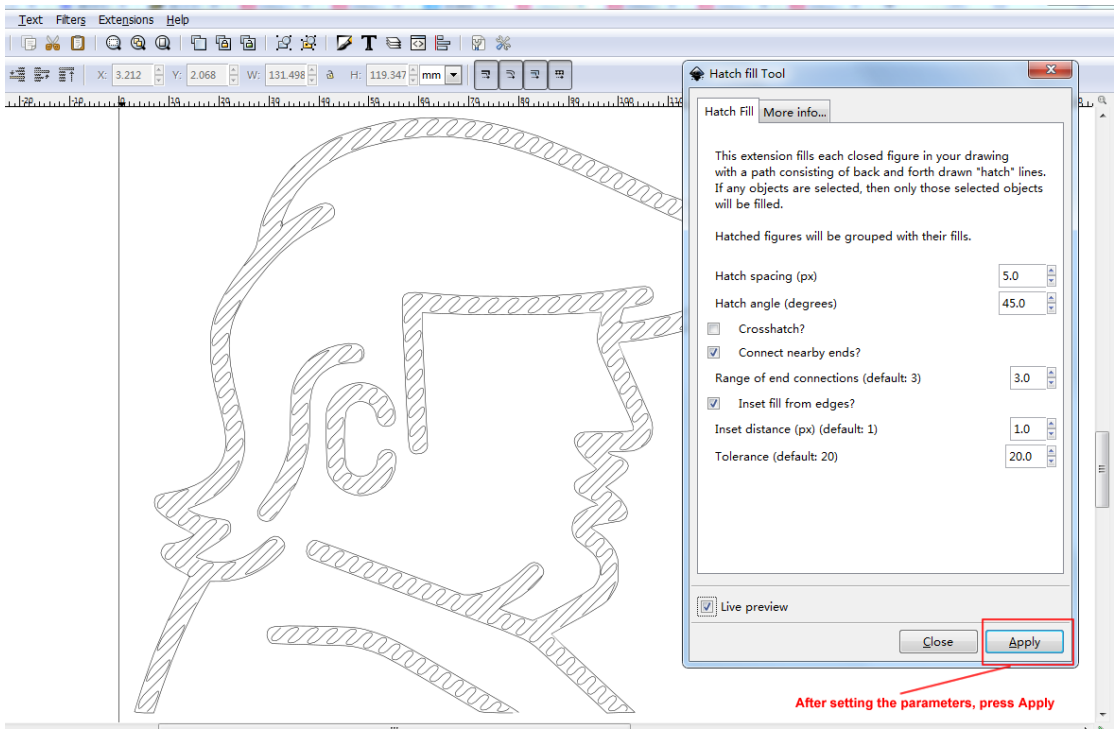

doesbot_0001.gcode - Universal Gcode Sender (Version 2.1.7)
File Edit Machine Program Tools Visualizer Window Help
Now you can use UGS software to send and run G-code
Toolbox
Macros
Jog Con...
doesbot_0001.gcode
Source
Visualizer
Z+
Y-
121.07 mm
133.18 mm
Step size XY 10 Millimet...
Step size Z 1 Larger
Feed rate 100 Smaller
Step size: 10.0mm
doesbot_0001.gcode
IDLE
Loaded File: 427 rows
1:1
INS Windows (CRLF)

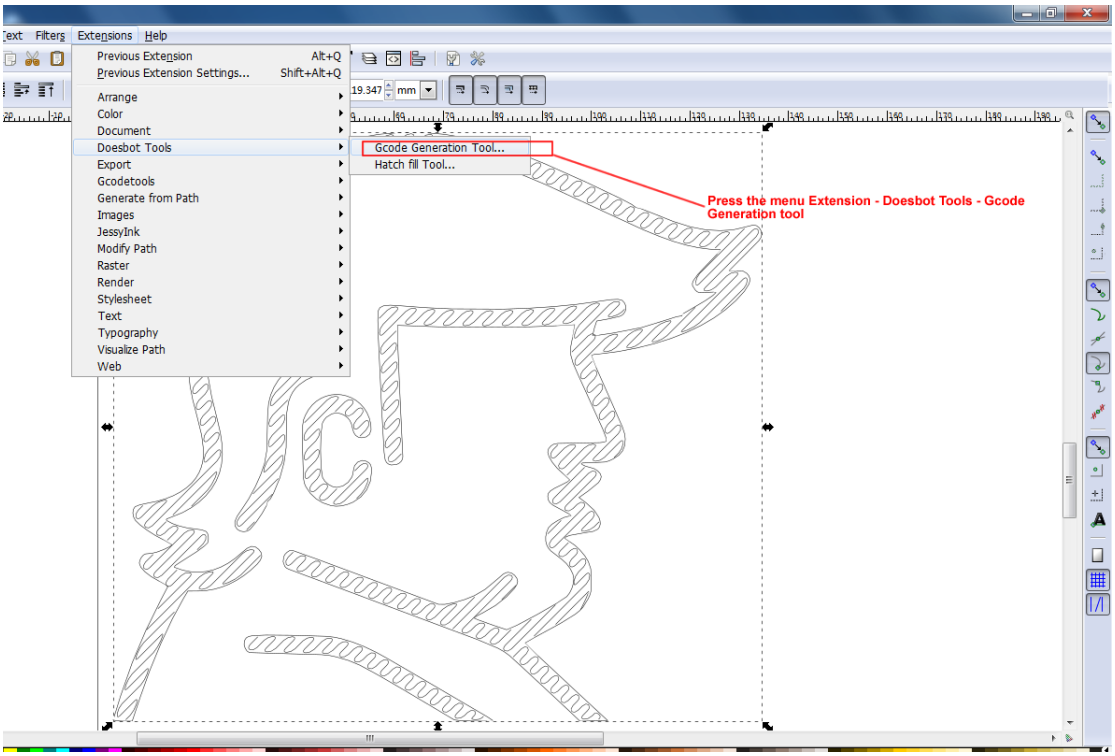
4. Fill image g code generation

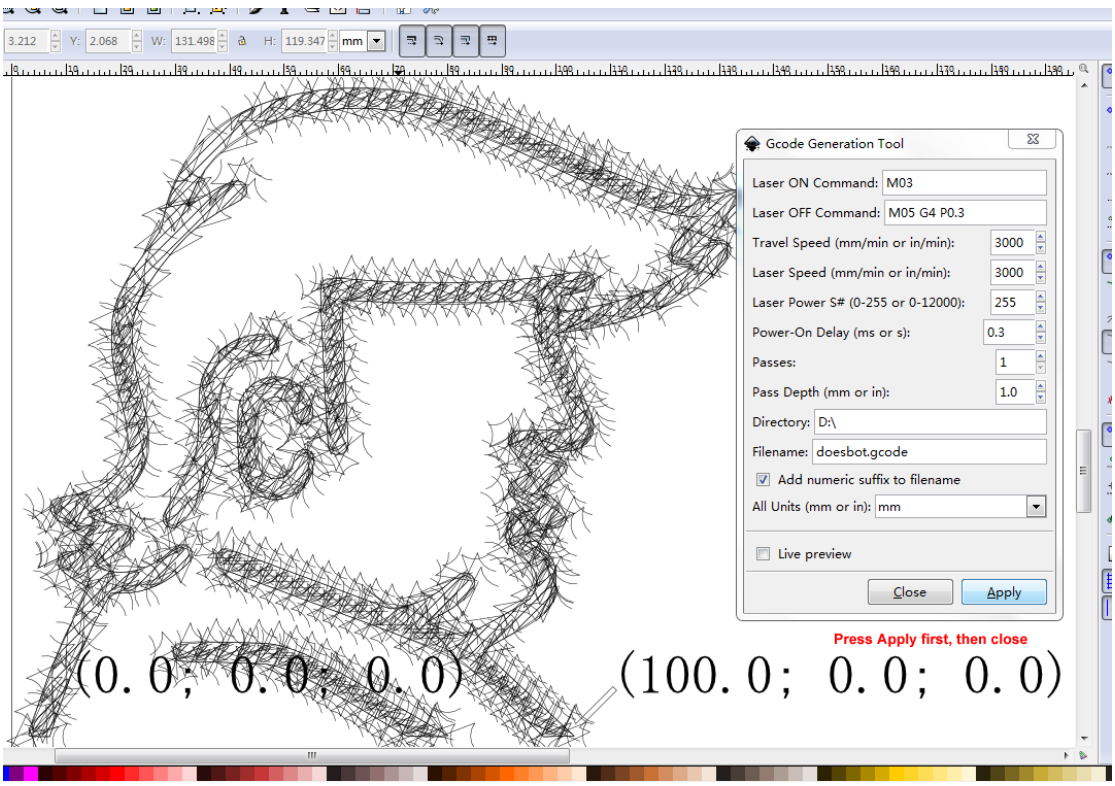


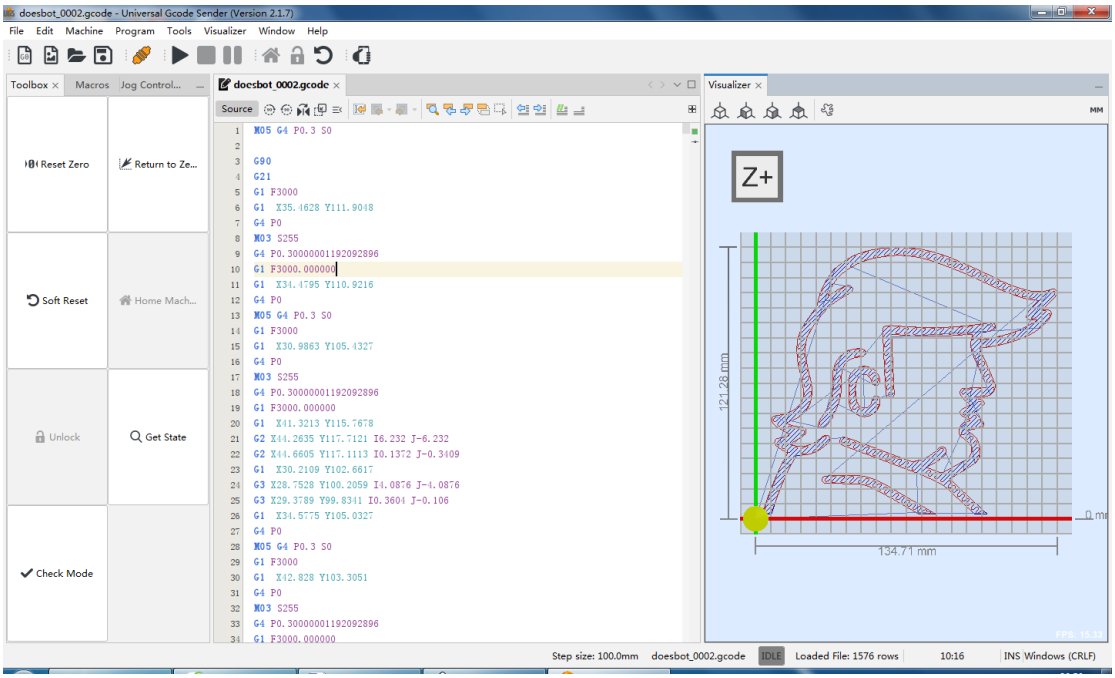



Text Filters Extensions Help
Previous Extension Alt+Q
Previous Extension Settings... Shift+Alt+Q
Arrange
Color
Document
Doesbot Tools
Export
Gcodetools
Generate from Path
Images
JessyInk
Modify Path
Raster
Render
Stylesheet
Text
Typography
Visualize Path
Web
Gcode Generation Tool...
Hatch fill Tool...
Then press the menu Extension - Doesbot Tools - Hatch fill Tool


Text Filters Extensions Help
Hatch fill Tool
Hatch Fill
More info...
This extension fills each closed figure in your drawing with a path consisting of back and forth drawn "hatch" lines. If any objects are selected, then only those selected objects will be filled.
Hatched figures will be grouped with their fills.
Hatch spacing (px) 5.0
Hatch angle (degrees) 45.0
Crosshatch?
Connect nearby ends?
Range of end connections (default: 3) 3.0
Inset fill from edges?
Inset distance (px) (default: 1) 1.0
Tolerance (default: 20) 20.0
Live preview
Close
Apply
After setting the parameters, press Apply


Text Filters Extensions Help
Previous Extension Alt+Q
Previous Extension Settings... Shift+Alt+Q
Arrange
Color
Document
Doesbot Tools
Export
Gcodetools
Generate from Path
Images
JessyInk
Modify Path
Raster
Render
Stylesheet
Text
Typography
Visualize Path
Web
Gcode Generation Tool...
Hatch fill Tool...
Press the menu Extension - Doesbot Tools - Gcode Generation tool


Gcode Generation Tool
Laser ON Command: M03
Laser OFF Command: M05 G4 P0.3
Travel Speed (mm/min or in/min): 3000
Laser Speed (mm/min or in/min): 3000
Laser Power S# (0-255 or 0-12000): 255
Power-On Delay (ms or s): 0.3
Passes: 1
Pass Depth (mm or in): 1.0
Directory: D:\
Filename: doesbot.gcode
Add numeric suffix to filename
All Units (mm or in): mm
Live preview
Close
Apply
Press Apply first, then close
(0. 0; 0. 0; 0. 0)
(100. 0; 0. 0; 0. 0)



Last update 4/28/2025

contact admin@doesbot.com